

Experiment No.09

Aim: CI/CD Deployment with GitHub Actions + Render/Vercel

Source Code:

Steps:

CI/CD with Render

Goal: Automatically deploy to Render when code is pushed to GitHub (**main** branch)

Step-by-Step Instructions:

1. Deploy your app on Render manually (first time)

- Go to Render
- Create a new Web Service or Static Site
- Connect your GitHub repo OR set up manually
- Wait until Render finishes the first deploy.

2. Create a Deploy Hook on Render

1. Go to your service's Settings
2. Scroll down to Deploy Hooks
3. Click "Add Deploy Hook"
4. Copy the Deploy Hook URL (you'll need this in the next step)

3. Add the Deploy Hook URL as a GitHub Secret

1. Go to your GitHub repository
2. Navigate to: **Settings > Secrets and Variables > Actions**
3. Click "New repository secret"
4. Name: **RENDER_DEPLOY_HOOK_URL**
Value: (paste the hook URL you copied)

4. Add GitHub Actions Workflow

1. In your project root, create the directory: **.github/workflows**
2. Inside it, create a file named: **deploy.yml**
3. Paste the following code:

```
# .github/workflows/deploy.yml
name: Deploy to Render
on:
```

```

push:
  branches:
    - master
jobs:
  deploy:

    runs-on: ubuntu-latest
    steps:
      - name: Trigger Render Deploy Hook
        run: |
          curl -X POST ${{ secrets.RENDER_DEPLOY_HOOK_URL }}

```

Now every time you push to the `main` branch, GitHub Actions will trigger a new deployment on Render

Output:

Push updated code to git

```

• PS D:\VESIT\SEM_V\FSD\FSD_PROGRAMS\GroupChatApp> git add .
• PS D:\VESIT\SEM_V\FSD\FSD_PROGRAMS\GroupChatApp> git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   frontend/package-lock.json

• PS D:\VESIT\SEM_V\FSD\FSD_PROGRAMS\GroupChatApp> git commit -m " Final Commit "
[master 3a2a9b5] Final Commit
1 file changed, 6 insertions(+)
create mode 100644 frontend/package-lock.json

```

```

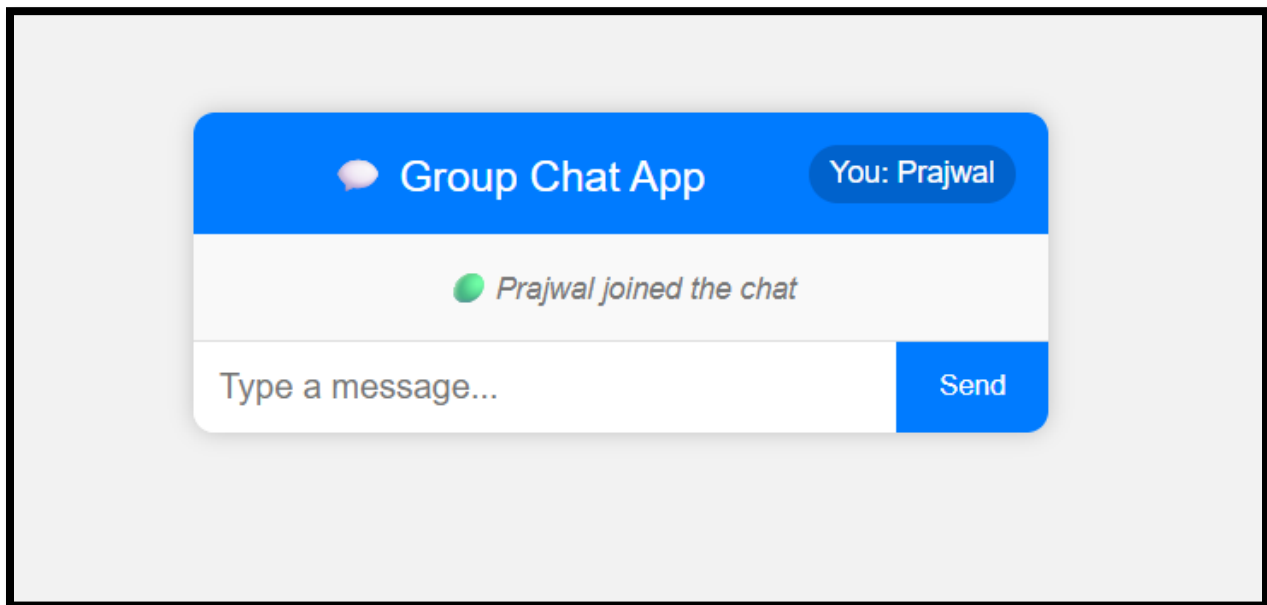
• PS D:\VESIT\SEM_V\FSD\FSD_PROGRAMS\GroupChatApp> git push -u origin master
Enumerating objects: 1109, done.
Counting objects: 100% (1109/1109), done.
Delta compression using up to 12 threads
Compressing objects: 100% (829/829), done.
Writing objects: 100% (1109/1109), 1.61 MiB | 1.45 MiB/s, done.
Total 1109 (delta 207), reused 1103 (delta 205), pack-reused 0
remote: Resolving deltas: 100% (207/207), done.
To https://github.com/Prajwal23k/GroupChat.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

```

On web services logs on render

```
Oct 6 11:20:38 PM ③ INFO      ==> Deploying...
Oct 6 11:20:43 PM ③ INFO      v8gvh  ● New user connected
Oct 6 11:20:57 PM ③ INFO      x6lcn  ==> Running 'npm start'
Oct 6 11:20:58 PM ③ INFO      x6lcn
Oct 6 11:20:58 PM ③ INFO      x6lcn  > backend@1.0.0 start
Oct 6 11:20:58 PM ③ INFO      x6lcn  > node server.js
Oct 6 11:20:58 PM ③ INFO      x6lcn
Oct 6 11:21:00 PM ③ INFO      x6lcn  🚀 Server running on http://localhost:5000
Oct 6 11:21:09 PM ③ INFO      ==> Your service is live 🎉
Oct 6 11:21:09 PM ③ INFO      ==>
Oct 6 11:21:09 PM ③ INFO      ==> //////////////////////////////////////
Oct 6 11:21:09 PM ③ INFO      ==>
Oct 6 11:21:09 PM ③ INFO      ==> Available at your primary URL https://g
Oct 6 11:21:09 PM ③ INFO      ==>
Oct 6 11:21:09 PM ③ INFO      ==> //////////////////////////////////////
Oct 6 11:21:22 PM ③ INFO      x6lcn  ● New user connected
```

After push on git



Conclusion:

In this setup, we integrated GitHub Actions with Render using a Deploy Hook to enable automatic deployment on every push to the `main/master` branch. This creates a simple and secure CI/CD pipeline, ensuring your app is always up to date without manual deployment.